Track B: Vision-guided pick with a UR3 in MuJoCo

Own project. Position-based visual servoing (PBVS) with a state-space LQG joint controller and a Kalman observer. Source: `track_b_vision_pick/`. All numbers below are generated from `results/`.

1. Summary

A UR3 (6 DOF, nominal DH kinematics) with a parallel-jaw gripper and a wrist-mounted RGB-D camera picks a red 4 cm cube from a tabletop. The cube position, yaw, mass and friction are randomised in every trial. The system calibrates the camera (intrinsics and hand-eye), localises the cube from depth, and approaches it with a two-stage PBVS. The joint controller is an LQG controller that acts on the estimated state. Joint velocities and the cube's position and velocity are unmeasured, and each is reconstructed by a Kalman filter.

In the nominal condition the grasp-and-lift succeeded in **50 of 50** trials. The median cube-centre error at the grasp plan was **0.59 mm** (95th percentile 1.19 mm), and the median yaw error was 0.08 degrees. Across the four stress conditions (Section 7), every trial succeeded.

2. Requirements mapping

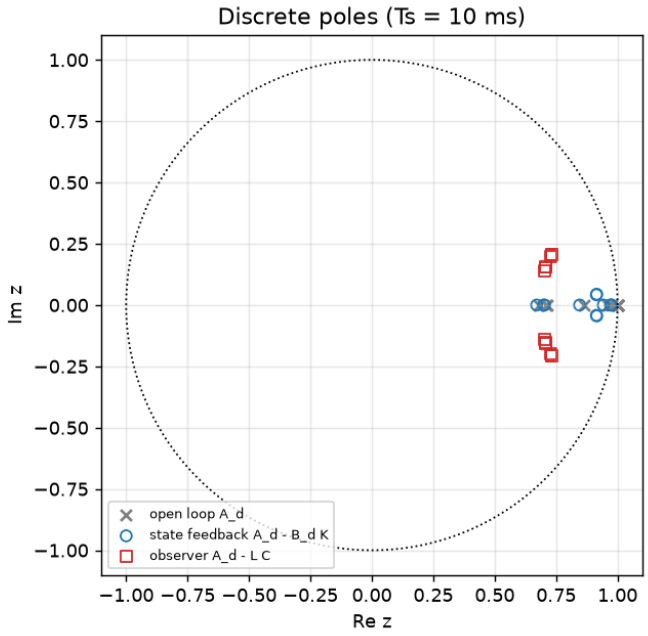
Requirement	Implementation
Camera calibration	40 checkerboard views were collected, with 36 retained after outlier rejection. Intrinsics by `calibrateCamera` (reprojection RMS 0.48 px). Hand-eye by `calibrateHandEye` (Park), then refined by minimising corner reprojection error jointly over the mount, the board pose and the intrinsics.
Depth-based localisation	HSV segmentation; back-projection of the depth image with calibrated K; transform to the base frame through encoder FK and the hand-eye; top-face extraction; yaw from the top-face minimum-area rectangle.
Approach planning	PBVS: pre-grasp from the estimate, re-observation at the pre-grasp view with a correction, descent, closing, grasp check, lift.
State-space model, controllability, observability	12-state arm model (positions, velocities). Ranks 12/12 and 12/12 for the design model; 12/12 and 12/12 for the full torque linearisation.
Calibrated exteroceptive sensor	Wrist RGB-D camera, calibrated intrinsics and hand-eye.
Kalman filter / observer	Joint observer (steady-state Kalman filter on encoder positions) and a constant-velocity Kalman filter on the cube position.
Feedback controller on estimated state	Computed torque plus LQR on `x_hat`, gain-scheduled per motion segment.
Robustness to randomised position	Cube (x, y, yaw, mass, friction) randomised in each trial; four stress conditions.

3. System and state-space model

The arm model is generated from the standard UR3 DH parameters (d_1=0.1519, a_2=-0.24365, a_3=-0.21325, d_4=0.11235, d_5=0.08535, d_6=0.0819 m), with approximate link masses. MuJoCo's forward kinematics match the DH model to numerical precision. The UR3 is not in MuJoCo Menagerie (only the UR5e is), so the links are capsules, not the UR meshes.

Plant state `x = [q; qdot]` (12 states), input `nu` (joint acceleration command), measurement `y = q` (encoders). Computed torque cancels the Coriolis and gravity terms, so the design model is `x' = A x + B nu`, with `A = [[0, I], [0, -M^-1 D]]` and `B = [[0], [I]]`. M is the mass matrix at the operating point and D the joint damping. This is discretised with a zero-order hold at Ts = 10 ms (the course's Week 5 method). The full torque-input linearisation, including the gravity gradient, is also computed from MuJoCo for the analysis.

Results at the scan configuration: both models are controllable and observable with the encoder output. The open-loop full linearisation has one unstable pole, with |lambda|max = 1.032, from gravity. The closed-loop poles lie at |z| <= 0.975 and the observer poles at |z| <= 0.761. The controllability and observability condition numbers are 44.5 and 130.4.



4. Calibration

The checkerboard (9 x 6 inner corners, 2 cm squares) sits on the table. Arm poses are sampled around it with tilt (±0.3 rad) and distance (0.22-0.42 m). Encoder readings are averaged over 50 samples per pose. Views whose error is an outlier are rejected. Against the simulator's known values (used only for reporting): focal length 582.3 px vs 579.4 px (true), principal point (321.8, 238.3) px vs (320.0, 240) px, and hand-eye translation error 2.05 mm with rotation error 0.22 degrees.

5. Estimation and control

Joint observer (unmeasured velocities)

The encoders measure only positions. A steady-state Kalman filter (Q from a 3 rad/s² acceleration disturbance, R from a 2 mrad encoder noise) reconstructs the joint velocities, which the controller uses for feedback.

The Kalman observer combines encoder position measurements with the mathematical state-space model to estimate the robot's unmeasured joint velocities. This allows the controller to use both position and velocity estimates when calculating its control input. The showcase graph in Section 7 compares the true joint velocities (solid lines) with the Kalman estimates (dotted lines), demonstrating how closely the observer follows the actual motion.

Target Kalman filter

The state is the cube centre and velocity. Each camera view is one position measurement, with a covariance that grows with the square of the range and a 2.5 mm lateral floor. The velocity is unmeasured. The filter is initialised from the first view and fuses the later views at the scan and pre-grasp poses.

Controller

LQR with Q = diag(400 I, 4 I) and R = 0.04 I, designed on the Ts = 10 ms model at each motion segment's start configuration (gain scheduling). Command: `tau = bias(x_hat) + M(q_op) * (q_ref_dd + M^-1 D q_ref_d - K (x_hat - x_ref))`. Joint trajectories are quintic (minimum-jerk) between IK waypoints.

6. Pick procedure

1. Move from the rest pose to the scan pose (tool pointing down, tool-centre point 0.25 m above the robot base). Capture two views.
2. Plan the pre-grasp 12 cm above the estimated cube centre. The gripper yaw is the cube yaw mod 90 degrees, closest to the current wrist.
3. PBVS correction: at the pre-grasp, capture two more views. If the refined grasp plan moved more than 4 mrad in joint space, re-plan and move.
4. Descend to the cube centre, close the gripper, and check the finger closure (a grasp gives about 10 mm; no cube gives 20 mm).
5. Lift 15 cm above the centre. Success is grasp detected and the cube above table by more than 5 cm (scored from truth after the trial).

The approach is hierarchical PBVS. Vision feeds back at the pre-grasp view only, not continuously during the descent. IBVS was not used.

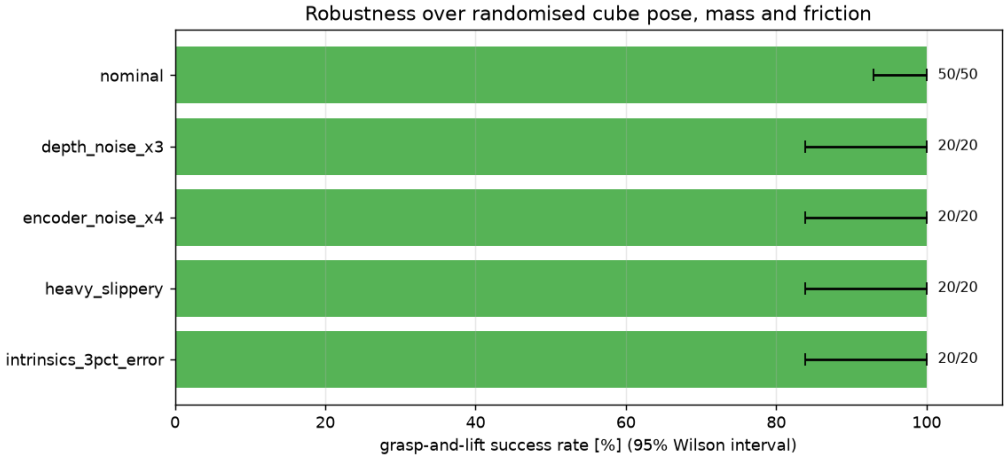
7. Experiments

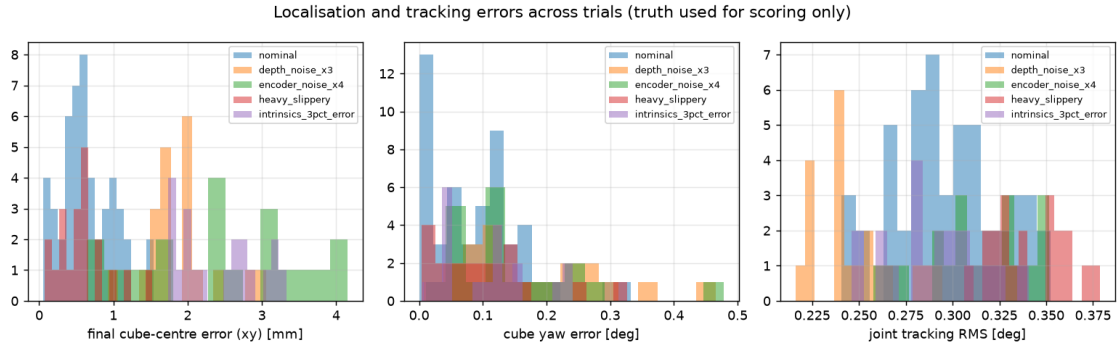
Each trial draws the cube position uniformly over x in [0.24, 0.38] m and y in [-0.14, 0.14] m, with yaw in ±45°. Mass and friction are

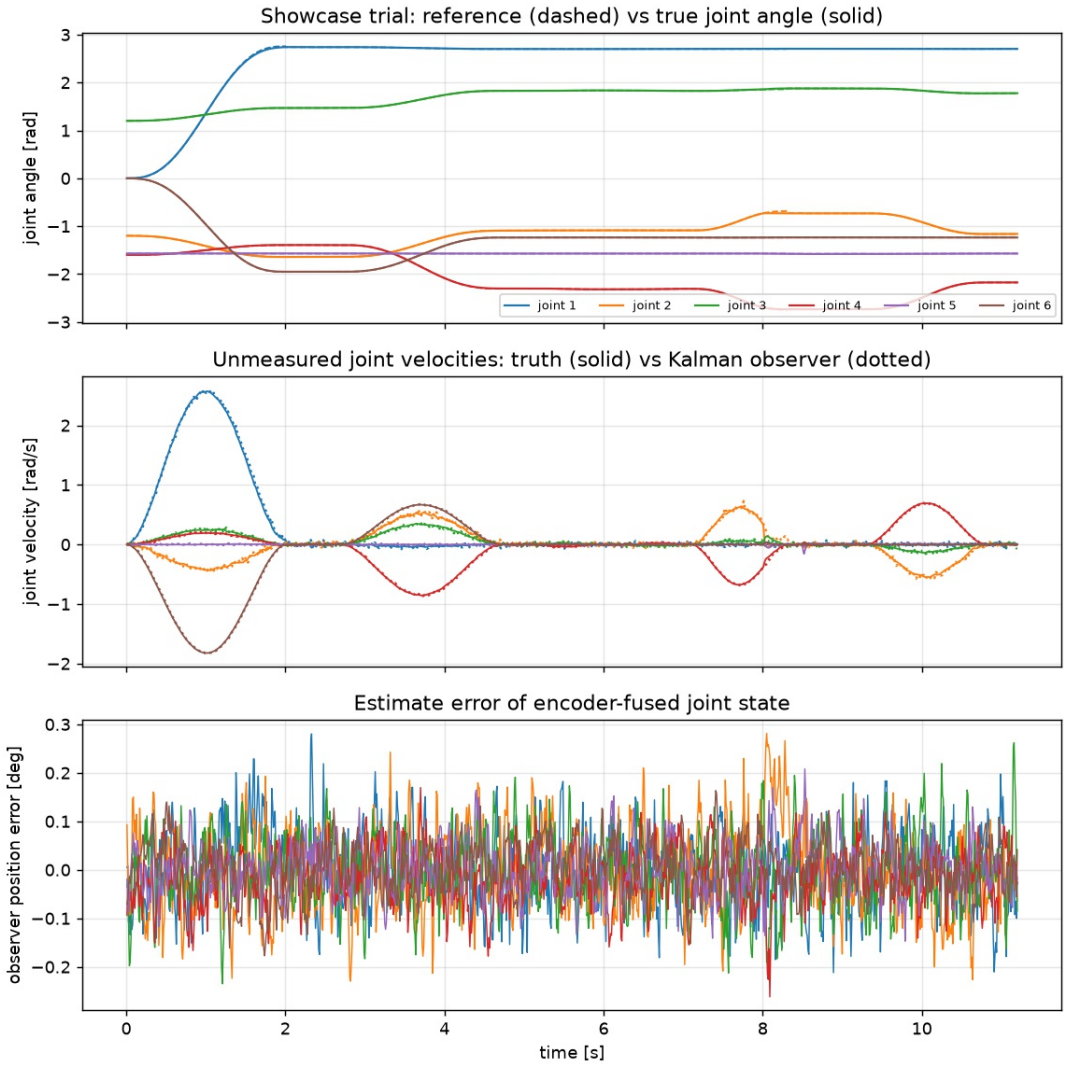
drawn from the ranges listed under each condition. Sensor noise: depth noise standard deviation 1 mm + 3 mm × Z^2 (Z in metres); encoders 2 mrad per sample. Ground truth is used only after each trial, for scoring.

Condition	Success	Success % [95% CI]	Cube xy error, median / p95 (mm)	Yaw error, median / p95 (deg)	Joint tracking error RMS, median (deg)	Velocity-observer RMS, median (rad/s)
nominal	50/50	100.0% [92.9, 100.0]	0.59 / 1.19	0.08 / 0.21	0.294	0.026
depth_noise_x3	20/20	100.0% [83.9, 100.0]	1.84 / 2.89	0.13 / 0.35	0.241	0.023
encoder_noise_x4	20/20	100.0% [83.9, 100.0]	2.40 / 4.14	0.12 / 0.27	0.314	0.077
heavy_slippery	20/20	100.0% [83.9, 100.0]	0.54 / 1.24	0.09 / 0.26	0.338	0.027
intrinsics_3pct_error	20/20	100.0% [83.9, 100.0]	2.38 / 3.17	0.11 / 0.23	0.291	0.026

Conditions: nominal (mass 30-100 g, friction 0.7-1.2, 50 trials); depth noise x3 (20 trials); encoder noise x4, i.e. 8 mrad per sample (20 trials); heavy and slippery (mass 100-200 g, friction 0.35-0.5, 20 trials); 3% focal-length error in the controller's intrinsics (20 trials). Success is grasp verified by finger closure and the cube lifted more than 5 cm; the 95% interval is a Wilson interval. Tracking RMS represents the root-mean-square difference between the reference and actual joint angles during the robot's motion. It is reported in degrees, with the table showing the median across trials. Although all 130 simulated trials were successful, this does not guarantee equivalent performance on physical hardware. Real-world performance would require further testing with actual sensors, calibration errors, mechanical dynamics and environmental disturbances.







8. Discussion and limitations

- The UR3 is modelled from DH parameters and approximate masses, with capsule geometry. The gripper is a simplified parallel jaw. Results should transfer to a real UR3 only after identification of its mass and friction parameters.
- The camera is an ideal pinhole model with no lens distortion and no rolling shutter, and the depth noise is a simple range-dependent model. Real depth sensors also have multipath and edge artefacts.
- The hand-eye and intrinsic calibrations are estimated from simulated images. The 3% focal-length condition in Section 7 tests the sensitivity to calibration error.
- The grasp is a top-down pinch on a cube of known size. The cube's size is a model assumption: the estimator sets the centre to the top-face height minus the known half-size.
- Gain scheduling is done once per segment, so the observer and controller are linear only locally. There is no integral action, and the steady-state error is handled by the computed-torque feedforward.
- The vision-based correction is applied at the pre-grasp view only. A continuous image-based loop during the descent would be the natural next step.